16.07 Dynamics Problem Set 3

Due Wednesday, October 14

Problem 1: Mass Properties of a T-Shaped Body

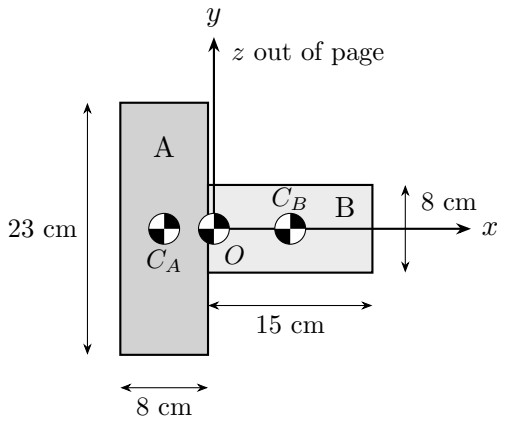


Figure 1: Top view of the T-shaped body, made of box A (the crossbar) and box B (the stem). Both boxes are 5 cm thick in the z direction, with their mid-planes at $z = 0$. The body-frame origin O is at the center of mass of the T-shaped body, and C_A and C_B mark the centers of mass of the two boxes.

The rigid T-shaped body in Fig. 1 is made of two rectangular boxes. We'll describe it in the body frame shown, whose origin O is at the center of mass of the T-shaped body: x runs along the stem (the body's axis of symmetry), y runs along the crossbar, and z completes a right-handed frame, normal to the plane of the T. When the body lies flat, z is vertical and rotation about z is *yaw*. We'll assume uniform density, so the centers of mass of the boxes, C_A and C_B, are at their geometric centers. The mass, dimensions, and moments of inertia of each box are given in Table 1. Throughout this problem set, we'll work in centimeter–gram–second (CGS) units. Each box's moments of inertia are about its own center of mass, along axes parallel to the body frame, and each box's products of inertia about its own center of mass (i.e., the off-diagonal elements of the inertia matrix) are zero.

Box	Mass (g)	Dimensions $x \times y \times z$ (cm)	Moments of inertia about own CoM $(J_{xx},\ J_{yy},\ J_{zz})$ $(10^3\ \mathrm{g\,cm^2})$
A (crossbar)	588.8	$8 \times 23 \times 5$	(27.18, 4.367, 29.10)
B (stem)	384.0	$15 \times 8 \times 5$	(2.848, 8.000, 9.248)

Table 1: Mass properties of the two boxes. Moments of inertia are about each box's own CoM, along axes parallel to the body frame.

(a) **Mass:** Find the total mass m of the T-shaped body.

(b) **Center of mass:** Find the position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ from the center of mass O of the T-shaped body to the box centers of mass C_A and C_B, respectively, expressed in the body frame. (You'll first need to locate O.)

(c) **Inertia matrix:** Using $\mathbf{r}_1$, $\mathbf{r}_2$, and the parallel axis theorem, find the inertia matrix $\mathbf{J}$ of the T-shaped body about its center of mass O, expressed in the body frame.

Problem 2: The Bifilar Pendulum

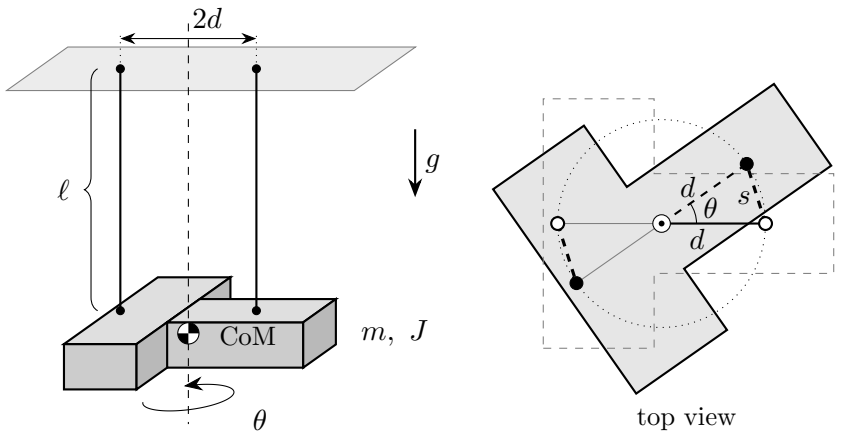


Figure 2: The T-shaped body from Problem 1 hung as a bifilar pendulum. Left: oblique view at rest; the strings attach to the top of the body on its x axis, a distance d on either side of the CoM. Right: top view with the body twisted by θ about the vertical axis through its center of mass (pointing out of the page). Open circles mark where the strings attach to the ceiling (fixed); filled circles mark where they attach to the body. The thick dashed segments of length s are the horizontal projections of the strings. Drawings are not to scale.

A bifilar ("two-string") pendulum is a classic technique for measuring the moment of inertia of an object. Early NACA engineers used it to measure the yaw moment of inertia of full-scale airplanes, and it is still widely used today for drones, small satellites, and aircraft components. In this problem, you'll derive the equation of motion for a general bifilar pendulum, then use the T-shaped body from Problem 1 as a working example.

A rigid body of mass m is suspended from the ceiling by two massless, inextensible strings, each of length ℓ. The strings attach to the ceiling at two points a horizontal distance $2d$ apart, and to the body at two points that are also $2d$ apart, so the strings hang vertical and parallel at rest. The body hangs with its center of mass (CoM) directly below the midpoint between the strings, as shown in Fig. 2. In our working example, the T-shaped body hangs flat, with its body z axis vertical, and the strings attach to its top surface at two points on its x axis, a distance d on either side of its CoM. Let J be the (unknown) moment of inertia of the body about the vertical axis through its CoM, and let θ be the angle the body is twisted about that axis, measured from its rest position.

When the body is twisted, the strings tilt and the body must rise slightly, so gravity provides a restoring torque. By measuring the period of small twisting oscillations, we can infer J. We'll make the following assumptions throughout this problem:

- The system is frictionless, and air resistance is negligible.
- The strings remain taut at all times.
- The body only twists: its CoM stays on the vertical axis (no side-to-side swinging) and the body stays level. The system then has a single generalized coordinate, θ, and the height of the body is determined by θ through the string-length constraint.

We'll use T for kinetic energy, U for potential energy, $\mathcal{L} = T - U$ for the Lagrangian, and P for the period of oscillation.

Equations of Motion

(a) **Geometry and energies:** Using the top view in Fig. 2, find the horizontal displacement s between the top and bottom ends of each string as a function of d and θ. Then use the fixed string length to find the vertical displacement $h(\theta)$ of the body. Write the kinetic energy T (including both the rotation of the body about the vertical axis and the vertical translation of its CoM) and the gravitational potential energy U, and combine them to obtain the Lagrangian $\mathcal{L}(\theta, \dot{\theta})$.

(b) **Small-angle expansion:** Taylor expand the Lagrangian to *second order* in the small quantities θ and $\dot{\theta}$ about the equilibrium $\theta = \dot{\theta} = 0$. Which terms survive? Explain why the kinetic energy of the body's vertical translation no longer appears.

(c) **Linearized equation of motion:** Apply the Euler–Lagrange equation to your quadratic Lagrangian to derive the linearized equation of motion, and write it in the standard form

$$\ddot{\theta} + \omega^2 \, \theta = 0.$$

(d) **Natural frequency and period:** Write the natural frequency ω (in rad/s) and the period P (in seconds) of small oscillations in terms of m, g, d, ℓ, and J. Using your values of m and $J = J_{zz}$ from Problem 1, $g = 981$ cm/s^2, $\ell = 200$ cm, and $d = 7.5$ cm, compute the natural frequency ω and the period P you would expect to measure.

(e) **Measuring the inertia:** Solve your result from part (d) for the moment of inertia J as a function of the measured period P and the other system parameters (analytically). We will use this expression in the lab to find J.